

Handcrafted Feature Optimisation and Classifier Benchmarking for Brain Tumor MRI Classification

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Abstract—Brain tumor classification from magnetic resonance imaging (MRI) supports triage of glioma, meningioma, and pituitary lesions. Most studies apply default parameters to handcrafted texture descriptors or couple exhaustive grid search to a single classifier, introducing optimisation bias and high compute cost. We present a two-phase framework on a public MRI dataset of 3,064 T1-weighted contrast-enhanced axial slices across three naturally imbalanced tumor classes (glioma, meningioma, pituitary): Phase 1 selects GLCM, LBP, DWT, and HOG hyperparameters using three classifier-free separability metrics (Fisher discriminant ratio, mutual information, Davies–Bouldin index) over 213 configurations; Phase 2 benchmarks eight classical models (SVM, KNN, random forest, XGBoost, LightGBM, logistic regression, MLP, ExtraTrees) with stratified 5-fold GridSearchCV on eight optimised feature compositions. The best cross-validated configuration pairs a polynomial-kernel SVM with the fused Full(opt) representation (accuracy = 0.920, macro-F1 = 0.920, Cohen’s κ = 0.88); optimized HOG features alone reach macro-F1 = 0.903 with SVM, and KNN on Full(opt) achieves macro-F1 = 0.900. Mean performance across classifiers confirms that fusion (Full(opt), mean F1 = 0.887) outperforms individual descriptor families, with gradient descriptors (D , mean F1 = 0.865) ranking highest among single-family sets. Full statistical validation (Cohen’s κ , bootstrap confidence intervals, McNemar, Wilcoxon, Friedman–Nemenyi) is reported for all 64 classifier–feature-set pairs. These results show that separability-guided handcrafted features with rigorous classical benchmarking achieve strong MRI classification without deep learning or GPU training.

Index Terms—Brain Tumor MRI, Handcrafted Features, Feature Optimization, Texture Analysis, GLCM, LBP, DWT, HOG, Machine Learning, Classifier Benchmarking, Statistical Validation

I. INTRODUCTION

Brain tumors represent one of the most critical neurological disorders, requiring accurate and timely diagnosis to improve treatment planning and patient outcomes. Magnetic Resonance Imaging (MRI) is widely used for brain tumor assessment because of its superior soft-tissue contrast and non-invasive nature. However, manual interpretation of MRI scans remains challenging due to the substantial variability in tumor appearance, shape, size, location, and texture. Consequently, computer-aided diagnosis (CAD) systems have emerged as

valuable tools for assisting radiologists in the classification of brain tumors [1].

Recent advances in artificial intelligence have significantly improved medical image analysis. Deep learning models, particularly Convolutional Neural Networks (CNNs), have achieved remarkable performance in brain tumor classification tasks [1]. Nevertheless, these approaches often require large annotated datasets, extensive computational resources, and specialized hardware for training. Furthermore, their decision-making process is frequently difficult to interpret, which may limit their adoption in clinical environments where transparency and explainability are important considerations [1].

In contrast, handcrafted feature-based approaches remain attractive for small- and medium-sized medical imaging datasets. Texture and structural descriptors such as the Gray-Level Co-occurrence Matrix (GLCM) [2], Local Binary Patterns (LBP) [3], the Discrete Wavelet Transform (DWT) [4], and the Histogram of Oriented Gradients (HOG) [5] provide interpretable representations of image characteristics while maintaining relatively low computational requirements. Numerous studies have demonstrated that carefully designed handcrafted features combined with classical machine learning classifiers can achieve competitive performance in brain tumor classification [6]–[9].

Despite their effectiveness, most existing studies rely on default descriptor parameters or optimize feature extractors using classifier-dependent criteria [6], [7], [9]. In such settings, the quality of the extracted features becomes intertwined with the behavior of a specific classifier, making it difficult to determine whether performance improvements originate from the feature representation itself or from the learning algorithm. Moreover, many studies evaluate only a limited number of classifiers and rarely provide comprehensive statistical analyses to verify the significance of observed performance differences [1], [6].

To address these limitations, this work proposes a two-phase framework for brain tumor MRI classification. In the first phase, handcrafted feature extractors are configured using a classifier-independent optimization strategy based on feature separability. Three complementary separability metrics—Fisher Discriminant Ratio (FDR), Mutual Information (MI),

and Davies–Bouldin Index (DBI)—are employed to evaluate and rank candidate parameter configurations for GLCM, LBP, DWT, and HOG descriptors. This approach enables the selection of feature representations that maximize class discrimination before any classifier training is performed.

In the second phase, the optimized feature representations are used to conduct a comprehensive benchmarking study of eight classical machine learning classifiers, namely Support Vector Machine (SVM) [10], K-Nearest Neighbors (KNN), Random Forest (RF) [11], XGBoost [12], LightGBM [13], Logistic Regression (LR), Multi-Layer Perceptron (MLP), and Extra Trees (ET). The benchmarking process evaluates multiple feature combinations and employs a rigorous experimental protocol based on stratified cross-validation and hyperparameter optimization. In addition, statistical validation techniques are applied to assess the reliability and significance of the obtained results.

The main contributions of this paper can be summarized as follows:

- 1) A separability-guided methodology for classifier-independent optimization of handcrafted feature extractors in brain tumor MRI classification.
- 2) A systematic evaluation of GLCM, LBP, DWT, and HOG descriptor configurations using Fisher Discriminant Ratio, Mutual Information, and Davies–Bouldin Index.
- 3) A comprehensive benchmark of eight classical machine learning classifiers across multiple optimized feature representations.
- 4) An analysis of feature-group importance that identifies which descriptor families contribute most to classification performance.
- 5) A rigorous statistical validation framework incorporating Cohen’s kappa coefficient [14], bootstrap confidence intervals, McNemar’s test [15], the Wilcoxon signed-rank test, and Friedman–Nemenyi analysis [16].

The results demonstrate that separability-guided feature configuration combined with statistically validated classifier benchmarking can provide competitive performance for brain tumor MRI classification while maintaining the interpretability and computational efficiency associated with handcrafted feature engineering.

II. RELATED WORK

A. Brain Tumor MRI Classification

Brain tumor classification from Magnetic Resonance Imaging (MRI) has become an active research area due to its potential to support early diagnosis and clinical decision-making. Traditional computer-aided diagnosis systems typically rely on handcrafted feature extraction followed by machine learning classifiers [17], [18], whereas recent approaches increasingly employ deep learning architectures capable of automatically learning discriminative representations from image data [1], [19].

Classical machine learning approaches generally consist of three main stages: image preprocessing, feature extraction,

and classification. Handcrafted descriptors are used to capture relevant characteristics of tumor regions, and the extracted features are subsequently classified using algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forests (RF), or ensemble learning methods [20], [21]. These approaches are particularly suitable when training data are limited and when interpretability is an important requirement.

Deep learning methods have recently achieved remarkable results in medical image analysis. Convolutional Neural Networks (CNNs), transfer learning models, attention mechanisms, and transformer-based architectures have demonstrated high classification accuracy across various brain tumor datasets [1], [19]. However, these models often require large amounts of annotated data, extensive computational resources, and specialized hardware. Furthermore, their black-box nature may limit their interpretability in clinical settings [1], [19].

Consequently, handcrafted feature-based methods remain highly relevant, particularly for small- and medium-scale datasets where model transparency, computational efficiency, and reproducibility are important considerations.

B. Handcrafted Feature-Based Approaches

1) *Texture-Based Features:* Texture descriptors are among the most widely used handcrafted features in brain tumor classification. Gray-Level Co-occurrence Matrix (GLCM) features [2] capture spatial relationships between pixel intensities and provide statistical measurements such as contrast, homogeneity, correlation, and energy. Numerous studies have shown that GLCM descriptors are effective for distinguishing different tumor types due to their ability to characterize tissue heterogeneity [6], [7].

Local Binary Patterns (LBP) [3] constitute another popular texture descriptor. By encoding local intensity variations into binary patterns, LBP provides a compact representation of micro-texture information and exhibits robustness to illumination changes. Several MRI classification studies have reported improved discrimination performance when combining LBP with other texture descriptors [7], [9].

2) *Frequency-Based Features:* Frequency-domain representations are commonly employed to capture multi-scale image information. The Discrete Wavelet Transform (DWT) [4] decomposes an image into multiple frequency sub-bands, enabling simultaneous analysis of spatial and frequency characteristics. DWT features have been successfully applied in medical imaging due to their ability to capture fine structural details and texture variations associated with different tumor classes [22].

3) *Gradient-Based Features:* Histogram of Oriented Gradients (HOG) [5] is a widely used descriptor that characterizes local shape and edge information through gradient orientation distributions. HOG features have demonstrated effectiveness in numerous computer vision applications and have been adopted in medical image classification tasks to capture structural and morphological properties of tumors [9], [20].

4) *Feature Fusion Strategies*: Recent studies have increasingly combined multiple handcrafted feature families to improve classification performance [6]–[8]. The rationale behind feature fusion is that different descriptors capture complementary information. Statistical features provide global intensity characteristics, texture descriptors capture local patterns, wavelet features encode frequency information, and gradient-based descriptors represent structural details. Feature fusion has frequently been reported to outperform single-descriptor approaches [6], [8].

C. Feature Optimization in Medical Imaging

The performance of handcrafted features is highly dependent on the selection of descriptor parameters [22]. For example, GLCM performance is influenced by quantization levels, distances, and angular directions; LBP depends on radius and neighborhood size; DWT performance varies according to wavelet family and decomposition level; and HOG effectiveness is affected by cell size, block size, and orientation binning.

Most existing studies adopt default parameter settings or perform optimization using classification accuracy as the objective function [8], [9], [23]. While effective, classifier-dependent optimization introduces a dependency between the feature extraction process and the selected learning algorithm [8], [23]. As a result, optimized parameters may not generalize across different classifiers.

Alternative optimization strategies based on feature separability have received comparatively less attention. Separability metrics evaluate the discriminative quality of feature representations independently of any classifier and therefore offer a more objective assessment of descriptor configurations. Despite their theoretical advantages, such approaches remain underexplored in brain tumor MRI classification.

D. Machine Learning Classifiers for Brain Tumor Classification

A variety of machine learning algorithms have been applied to brain tumor classification [20], [21].

Support Vector Machines (SVMs) [10] are among the most frequently used classifiers due to their strong generalization capabilities and effectiveness in high-dimensional feature spaces. K-Nearest Neighbors (KNN) provides a simple yet effective non-parametric alternative. Ensemble methods such as Random Forests [11] and Extra Trees improve robustness through the aggregation of multiple decision trees.

More recently, gradient boosting algorithms including XGBoost [12] and LightGBM [13] have demonstrated excellent predictive performance across numerous biomedical classification tasks [23], [24]. Logistic Regression remains a strong baseline model because of its interpretability, while Multi-Layer Perceptrons (MLPs) provide nonlinear modeling capabilities without requiring deep architectures.

Although many studies report results for one or two classifiers, relatively few investigations perform large-scale benchmarking across a diverse set of machine learning models under identical experimental conditions [1], [20].

E. Research Gap and Positioning

The literature reveals three major limitations.

First, handcrafted feature extractors are frequently used with default or empirically selected parameters, while systematic optimization of descriptor configurations remains limited [7], [9].

Second, when optimization is performed, it is typically classifier-dependent, making it difficult to distinguish improvements attributable to feature quality from those resulting from classifier behavior [8], [23].

Third, existing studies often evaluate only a small number of classifiers and rarely perform rigorous statistical significance testing to validate performance differences [6], [25].

To address these limitations, this work introduces a separability-guided feature configuration framework that optimizes handcrafted descriptors independently of classifier performance. The resulting optimized feature representations are then evaluated through a comprehensive benchmarking study involving eight machine learning classifiers and multiple feature combinations. Finally, statistical validation techniques are employed to assess the significance and reliability of the observed results, providing a more rigorous evaluation than commonly reported in the literature [1], [25].

III. MATERIALS AND EXPERIMENTAL SETUP

A. Dataset

This study was conducted using a publicly available brain tumor Magnetic Resonance Imaging (MRI) dataset containing three tumor categories: Glioma, Meningioma, and Pituitary Tumor [26]. The dataset comprises 3,064 T1-weighted contrast-enhanced MRI slices collected from clinical examinations and has been widely used as a benchmark in brain tumor classification research.

The class distribution is naturally imbalanced, comprising approximately 1,426 glioma, 708 meningioma, and 930 pituitary images (imbalance ratio ≈ 2.0). Rather than artificially resampling the data, the native distribution is preserved and class imbalance is addressed through stratified data partitioning, class-weighted training where supported by the classifier, and macro-averaged evaluation metrics that assign equal importance to every class. Figure 1 illustrates representative examples from the three classes.

The classification task is formulated as a three-class supervised learning problem, with class labels 0, 1, and 2 corresponding to glioma, meningioma, and pituitary tumor, respectively.

B. Image Preprocessing

Medical images often exhibit variations in intensity distribution, image quality, and acquisition conditions. Therefore, a preprocessing pipeline was applied prior to feature extraction to enhance image consistency and improve descriptor robustness.

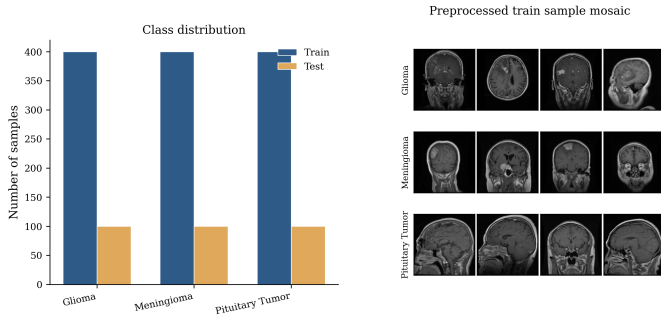


Fig. 1: Representative MRI slices from the three tumor classes (glioma, meningioma, and pituitary tumor).

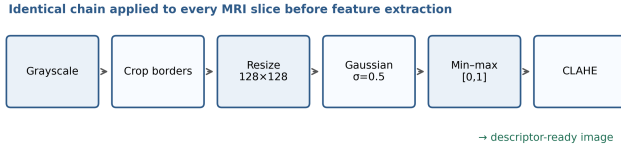


Fig. 2: Preprocessing pipeline applied to each MRI slice before feature extraction.

1) *Grayscale Conversion*: Since the MRI images contain limited color information and most handcrafted descriptors operate on intensity values, all images were represented in grayscale. This reduces computational complexity while preserving diagnostically relevant structural information.

2) *Image Resizing*: To ensure uniform feature extraction across the dataset, all images were resized to a fixed spatial resolution of 128×128 pixels using anti-aliased interpolation. Standardizing image dimensions prevents descriptor inconsistencies caused by varying image sizes and facilitates fair comparison between samples.

3) *Contrast Enhancement*: Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied (clip limit 0.02) to improve local contrast and enhance subtle tumor-related structures. Unlike conventional histogram equalization, CLAHE limits noise amplification while increasing the visibility of clinically relevant texture patterns.

4) *Intensity Normalization*: Pixel intensity values were normalized to the range $[0, 1]$ to reduce variations introduced by different acquisition settings and imaging devices, improving the stability and reproducibility of the extracted feature representations.

The preprocessing pipeline can therefore be summarized as: MRI image $\rightarrow$ grayscale $\rightarrow$ resizing $\rightarrow$ CLAHE enhancement $\rightarrow$ intensity normalization (Figure 2).

C. Overall Experimental Framework

The proposed framework consists of two sequential phases designed to separate feature optimization from classifier evaluation.

a) *Phase 1: Separability-Guided Feature Configuration*: The first phase identifies the most discriminative parameter

configurations for each handcrafted descriptor family. Candidate configurations are evaluated using classifier-independent separability metrics, allowing feature quality to be assessed without introducing bias from any particular learning algorithm. The descriptor families investigated are the Gray-Level Co-occurrence Matrix (GLCM), Local Binary Patterns (LBP), the Discrete Wavelet Transform (DWT), and the Histogram of Oriented Gradients (HOG). For each descriptor, a search space of parameter combinations is explored and ranked according to a composite separability score.

b) *Phase 2: Classifier Benchmarking*: The optimized descriptors obtained in Phase 1 are used to construct multiple feature representations, which are then evaluated using a diverse set of machine learning classifiers under identical experimental conditions. This two-phase design ensures that feature optimization and classifier benchmarking remain independent, enabling a more objective assessment of both components.

D. Feature Representation and Cleaning

From the optimized descriptors, eight feature representations were constructed: the individual families (A: first-order statistics; B: GLCM+LBP texture; C: DWT; D: HOG) and four fusions (A+B, B+C, A+B+C, and the complete set Full(opt)). Each representation is passed through a cleaning and standardization pipeline (variance filtering, correlation pruning, and training-set z-score normalization) before classification. The construction of these representations and the cleaning pipeline are detailed in Section V.

E. Machine Learning Classifiers

Eight machine learning algorithms spanning linear, distance-based, tree-based, ensemble, boosting, and neural-network paradigms were evaluated: SVM, KNN, Random Forest, Extra Trees, XGBoost, LightGBM, Logistic Regression, and a Multi-Layer Perceptron. Each algorithm and its hyperparameter search space is described in Section VI.

F. Hyperparameter Optimization

To ensure a fair comparison between classifiers, hyperparameter optimization was performed using grid search combined with stratified cross-validation. For each classifier, a predefined search space was explored, and the configuration achieving the highest macro-averaged F_1 score in cross-validation was selected. Macro- F_1 was adopted as the selection criterion so that model tuning is not dominated by the majority class. This procedure reduces the risk of underestimating classifier performance due to suboptimal parameter choices.

G. Evaluation Protocol

The dataset was divided into training and testing subsets using a stratified hold-out strategy (80%/20%) that preserves class proportions. Within the training set, model selection and hyperparameter tuning were performed using five-fold stratified cross-validation. Performance was reported using accuracy, together with macro-averaged precision, recall, and F_1 score. Macro-averaging ensures equal importance across

all tumor classes, which is appropriate given the imbalanced class distribution.

H. Statistical Validation

Beyond conventional performance metrics, statistical validation was conducted to assess the significance and robustness of observed performance differences. The following analyses were employed: Cohen’s kappa coefficient [14]; bootstrap 95% confidence intervals; McNemar’s test [15] for pairwise comparison of classifiers on the same test set; the Wilcoxon signed-rank test; and the Friedman rank test with Nemenyi post-hoc analysis [16] for comparison across multiple classifiers and feature sets. These procedures provide a rigorous framework for determining whether performance improvements are statistically meaningful rather than the result of random variation.

The methodology described in this section establishes a reproducible experimental framework for evaluating separability-guided handcrafted feature optimization and machine learning classifier benchmarking in brain tumor MRI classification.

IV. PHASE 1: SEPARABILITY-GUIDED FEATURE CONFIGURATION

A. Motivation

The performance of handcrafted feature-based classification systems depends heavily on the parameter configuration of the underlying feature extractors. Parameters such as quantization levels in the GLCM, neighborhood settings in LBP, decomposition levels in the DWT, and cell structures in HOG directly influence the discriminative quality of the extracted representations.

Most existing studies either employ default parameter values or optimize descriptors using classifier-dependent criteria such as classification accuracy. Although effective, classifier-dependent optimization introduces a coupling between feature extraction and classifier behavior, making it difficult to determine whether performance improvements arise from the quality of the feature representation or from the characteristics of a particular learning algorithm. To overcome this limitation, this work adopts a classifier-independent optimization strategy based on feature separability: candidate configurations are assessed according to their ability to separate tumor classes in the feature space before any classifier training is performed. The objective of Phase 1 is therefore to identify descriptor configurations that maximize class discrimination while remaining independent of any specific machine learning model.

B. Descriptor Search Space

Four handcrafted feature families were investigated. For each, a grid of candidate parameter configurations was generated from the training images and evaluated, yielding 144 GLCM, 36 LBP, 21 DWT, and 12 HOG configurations (213 in total).

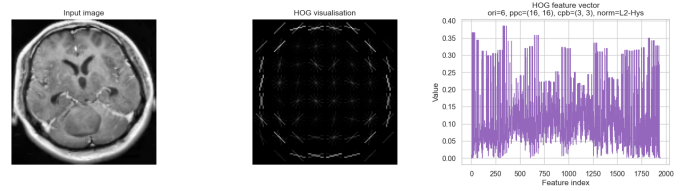


Fig. 3: Illustration of HOG descriptor computation on a preprocessed MRI slice.

1) *GLCM Search Space*: The Gray-Level Co-occurrence Matrix describes second-order texture statistics by modeling spatial relationships between pixel intensities. The explored parameters were the number of gray levels, the set of pixel distances, the set of angular directions, and matrix symmetrization. For each configuration, six texture properties—contrast, dissimilarity, homogeneity, energy, correlation, and angular second moment (ASM)—were extracted and averaged over all distance–angle pairs, producing a six-dimensional descriptor.

2) *LBP Search Space*: Local Binary Patterns characterize local texture structures by comparing neighboring pixels with a central pixel. The explored parameters were the number of sampling points P , the neighborhood radius R , and the encoding method (uniform, nri_uniform, and ror). The resulting pattern histograms were used as texture descriptors for separability evaluation.

3) *DWT Search Space*: The Discrete Wavelet Transform provides a multi-resolution representation of image content by decomposing an image into frequency sub-bands. The search space comprised seven wavelet families and three decomposition levels. Four statistical measures were extracted from each sub-band (approximation and detail coefficients), yielding $4(3L + 1)$ features for a decomposition of level L .

4) *HOG Search Space*: The Histogram of Oriented Gradients captures local shape and structural information through gradient orientation distributions. The explored parameters were the number of orientation bins, the cell size (pixels per cell), and the block size (cells per block); block normalization was fixed to L2-Hys. Each combination generated a different gradient-based representation (Figure 3).

C. Separability Metrics

To assess descriptor quality independently of classifier performance, three complementary separability metrics were employed. Prior to evaluation, each candidate feature matrix was standardized, and all metrics were computed on the training set only.

1) *Fisher Discriminant Ratio (FDR)*: The Fisher Discriminant Ratio relates inter-class separation to intra-class variability. For a single feature it is defined as

$$\text{FDR} = \frac{\sum_{c=1}^C n_c (\mu_c - \mu)^2}{\sum_{c=1}^C n_c \sigma_c^2}, \quad (1)$$

where C is the number of classes, n_c , μ_c , and σ_c^2 are the size, mean, and variance of class c , and μ is the global mean. The

score is averaged across all features; higher values indicate better class discrimination.

2) *Mutual Information (MI)*: Mutual Information measures the statistical dependency between a feature X and the class label Y ,

$$MI(X, Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}. \quad (2)$$

In practice it is estimated with a nearest-neighbor estimator ($k = 5$) and averaged across features; higher values indicate more class-relevant information.

3) *Davies–Bouldin Index (DBI)*: The Davies–Bouldin Index [27] evaluates cluster compactness and separation,

$$DBI = \frac{1}{C} \sum_{i=1}^C \max_{j \neq i} \left(\frac{S_i + S_j}{M_{ij}} \right), \quad (3)$$

where S_i and S_j denote intra-cluster dispersion and M_{ij} the distance between the centroids of classes i and j . To keep the computation efficient in high dimensions, the DBI is evaluated on a PCA-reduced projection (at most 20 components) of the standardized features. Unlike FDR and MI, lower DBI values indicate better separability.

D. Composite Separability Score

Because each metric captures a different aspect of class discrimination, a composite score was constructed as a unified ranking criterion. Within each descriptor family, metric values were first normalized to $[0, 1]$ across the candidate configurations. For FDR and MI,

$$M_{\text{norm}} = \frac{M - M_{\min}}{M_{\max} - M_{\min}}, \quad (4)$$

and for the DBI, where lower values are preferred,

$$DBI_{\text{norm}} = 1 - \frac{DBI - DBI_{\min}}{DBI_{\max} - DBI_{\min}}. \quad (5)$$

The composite score is the equally weighted average

$$\text{Score} = w_1 \text{FDR}_{\text{norm}} + w_2 \text{MI}_{\text{norm}} + w_3 \text{DBI}_{\text{norm}}, \quad (6)$$

$$w_1 = w_2 = w_3 = \frac{1}{3}.$$

Because normalization is performed *within* each descriptor family, the composite score ranks configurations of the same descriptor against one another; it should not be interpreted as an absolute, cross-descriptor measure of feature quality.

E. Optimization Procedure

The optimization procedure was performed independently for each descriptor family. For every candidate configuration, features were extracted from the training images; the FDR, MI, and DBI were computed; the metric values were normalized and combined into the composite score; and the configurations were ranked. The highest-ranked configuration for each family was retained for the subsequent experiments. The procedure can be summarized as: configuration generation $\rightarrow$ feature extraction $\rightarrow$ separability evaluation $\rightarrow$ composite scoring $\rightarrow$ ranking $\rightarrow$ optimal configuration selection.

Figure 4 visualizes the composite separability score across the explored parameter grids for all four descriptor families.

TABLE I: Optimal descriptor configurations selected in Phase 1. Composite scores are normalized within each descriptor family and are therefore not directly comparable across families.

Descriptor	Optimal parameters	Dim.	Score
GLCM	dist. {1, 2, 4, 8}, 1 angle (0°), 128 levels, asymmetric	6	0.7468
LBP	$P = 12$, $R = 1$, uniform	14	0.9906
DWT	db2, level 1	16	0.9837
HOG	6 orient., 16×16 cell, 3×3 block, L2-Hys	1176	0.9898

F. Optimal Descriptor Configurations

Table I reports the selected configuration for each descriptor family, together with its dimensionality and composite score. The best GLCM configuration combined four pixel distances at a single orientation with 128 gray levels and an asymmetric co-occurrence matrix. The optimal LBP descriptor used twelve uniform sampling points at radius one. For the DWT, the db2 wavelet at a single decomposition level offered the best separability score. The optimal HOG configuration used six orientation bins with 16×16 -pixel cells and 3×3 -cell blocks.

Figure 5 plots the separability metrics for the selected optimal configuration within each family.

G. Discussion of Phase 1 Results

The optimization results demonstrate that descriptor parameterization has a substantial impact on feature discriminability. Marked differences were observed between candidate configurations within each descriptor family, confirming that default parameter settings do not necessarily produce optimal representations for brain tumor MRI classification.

The proposed separability-guided framework offers several advantages. First, it enables an objective evaluation of descriptor quality without relying on classifier performance. Second, it reduces the risk of overfitting feature configurations to a specific learning algorithm. Third, it provides a systematic methodology for selecting feature extractors before classifier benchmarking is performed. It should be emphasized that the composite scores are a classifier-independent *selection* criterion computed on the training set, rather than an estimate of classification accuracy; the predictive value of each representation is assessed separately in Phase 2.

The optimized descriptors obtained in this phase constitute the foundation for the classifier benchmarking experiments presented in the next section. By ensuring that all classifiers operate on carefully selected feature representations, the subsequent comparison focuses on differences attributable to the classifiers themselves rather than to variations in feature quality.

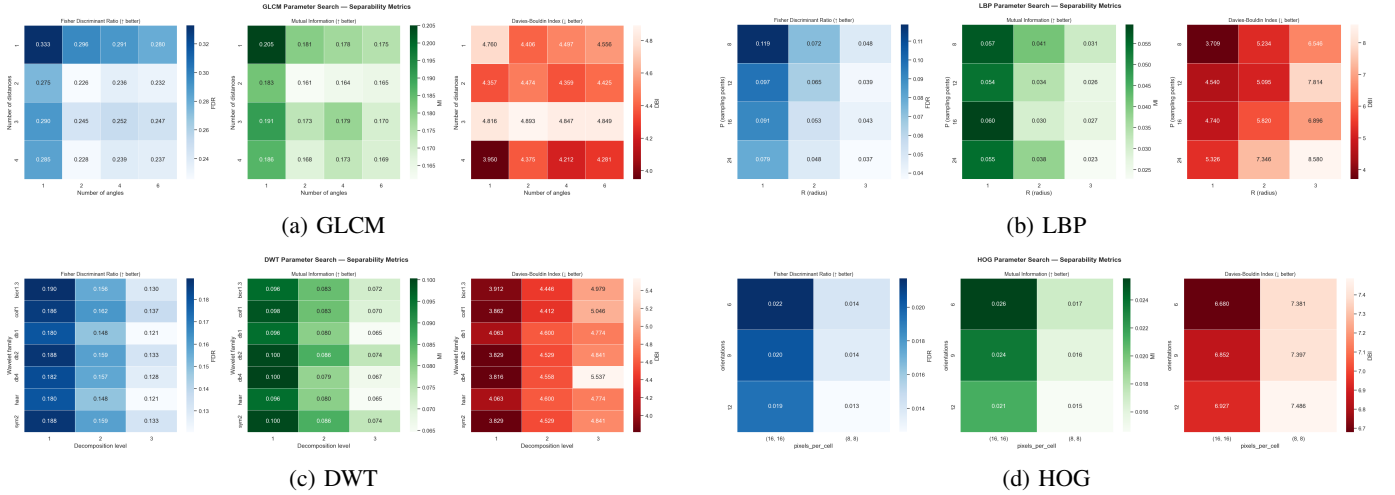


Fig. 4: Phase 1 parameter search: composite separability score across candidate configurations for each descriptor family.

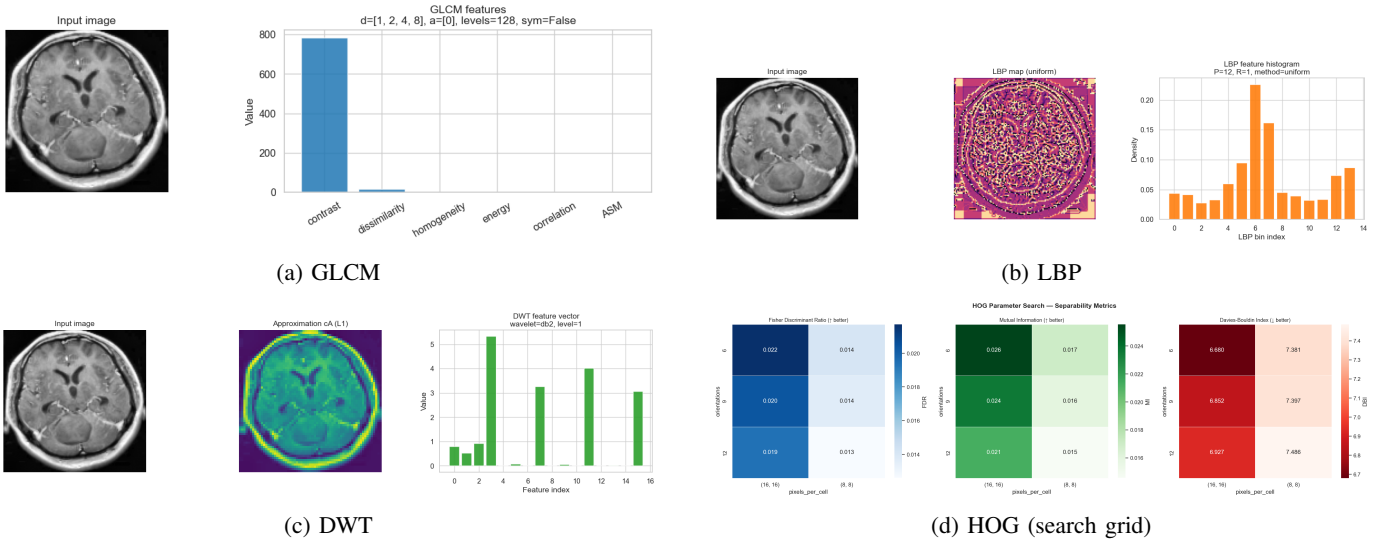


Fig. 5: Separability profiles for the optimal GLCM, LBP, and DWT configurations and the HOG parameter grid.

V. FEATURE REPRESENTATION CONSTRUCTION AND CLEANING

A. Overview

Using the optimal descriptor configurations from Section IV (Phase 1), multiple feature representations were constructed for classifier evaluation. This serves two purposes: to investigate the individual contribution of each feature family, and to assess whether combining complementary descriptors improves discrimination. The construction progressively integrates intensity statistics, texture patterns, frequency-domain information, and structural gradients.

B. Statistical Features (A)

First-order statistical features summarize the global intensity distribution of an MRI image. For each preprocessed image, eleven descriptors were extracted: the mean, standard deviation, and variance of the intensities; the histogram entropy

(Shannon entropy over 64 intensity bins); the skewness and kurtosis of the intensity distribution; and five percentiles (10th, 25th, 50th/median, 75th, and 90th). Together these capture the brightness level, spread, and shape of the intensity distribution associated with different tumor types. This set is denoted A .

C. Texture Features (B)

Texture is among the most informative cues in brain tumor MRI, since tumor tissues often exhibit distinct local patterns and structural heterogeneity. The texture representation concatenates the optimized GLCM and LBP descriptors. The GLCM component captures second-order spatial relationships through the six properties selected in Phase 1 (contrast, dissimilarity, homogeneity, energy, correlation, and ASM), while the LBP component encodes local micro-patterns as uniform binary signatures. The fused texture set is $B = \text{GLCM}_{\text{opt}} \oplus \text{LBP}_{\text{opt}}$, where $\oplus$ denotes concatenation.

TABLE II: Evaluated feature representations. Dimensionality is reported before cleaning; variance and correlation filtering subsequently reduce these counts.

Set	Components	Dim.
A	Statistical features	11
B	Optimized GLCM + LBP	20
C	Optimized DWT	16
D	Optimized HOG	1176
A+B	Statistical + Texture	31
B+C	Texture + Frequency	36
A+B+C	Statistical + Texture + Frequency	47
Full(opt)	Statistical + Texture + Frequency + Gradient	1223

D. Frequency-Domain Features (C)

Many tumor-related structures are discriminative across spatial frequencies. The optimized DWT decomposes each image into sub-bands corresponding to different frequency ranges and orientations, and statistical measures from the approximation and detail coefficients summarize both coarse and fine structures. This set is $C = \text{DWT}_{\text{opt}}$.

E. Gradient-Based Features (D)

Structural boundaries, edges, and shape cues help distinguish tumor classes. The optimized HOG descriptor characterizes these properties through local distributions of gradient orientations, focusing on the geometric and morphological aspects of tumor regions. This set is $D = \text{HOG}_{\text{opt}}$.

F. Feature Fusion Strategies

Because individual families describe different aspects of the image, combining them may improve discrimination through complementary information. Four fusion representations were constructed: $A+B$ (statistics and texture), $B+C$ (texture and frequency), $A+B+C$ (statistics, texture, and frequency), and $\text{Full}(\text{opt}) = A+B+C+D$, which incorporates all optimized descriptors. Table II summarizes the eight evaluated representations together with their raw (pre-cleaning) dimensionality.

G. Motivation for Feature Cleaning

Feature fusion increases dimensionality, which can introduce redundant and highly correlated variables, noise sensitivity, additional computational cost, and a greater risk of overfitting. A cleaning pipeline was therefore applied to each representation prior to classifier training.

H. Variance-Based Filtering

The first stage removes constant features, whose variance

$$\text{Var}(f) = \frac{1}{N} \sum_{i=1}^N (f_i - \bar{f})^2 \quad (7)$$

is zero across the training set (N samples, feature values f_i , mean $\bar{f}$). Such features carry no discriminative information and only increase model complexity.

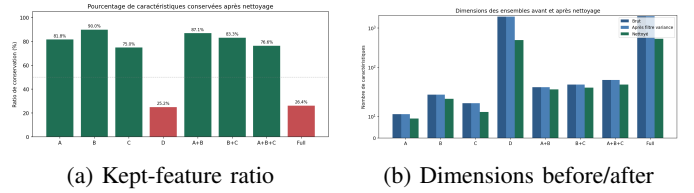


Fig. 6: Effect of variance and correlation filtering on feature-set dimensionality.

I. Correlation-Based Reduction

Even after variance filtering, many handcrafted descriptors remain highly correlated and convey nearly identical information, which can harm model stability. Pairwise Pearson correlations

$$\rho_{ij} = \frac{\text{Cov}(f_i, f_j)}{\sigma_i \sigma_j} \quad (8)$$

were computed, and for each pair with $|\rho_{ij}| > 0.95$ one feature was removed. This pruning is applied whenever the representation is of moderate dimensionality (below 5000 features), which holds for all eight sets considered here.

J. Standardization

The retained features were standardized by z-score normalization,

$$z = \frac{x - \mu}{\sigma}, \quad (9)$$

where μ and σ are estimated on the training set only and applied unchanged to the test set, preventing information leakage. Standardization ensures that features on different scales contribute comparably and is particularly important for distance- and margin-based classifiers such as KNN and SVM.

K. Final Feature Spaces

After variance filtering, correlation reduction, and standardization, the cleaned representations are discriminative (owing to separability-guided descriptor selection), compact (owing to filtering), and classifier-ready (owing to standardization). Figure 6 summarizes the dimensionality reduction achieved by the cleaning pipeline; high-dimensional sets containing HOG retain a smaller fraction of features owing to correlation pruning. Figure 7 visualizes class separability in the cleaned $\text{Full}(\text{opt})$ representation via t-SNE projection. These representations form the experimental basis for the classifier benchmarking study presented in the next section.

VI. PHASE 2: CLASSIFIER BENCHMARKING

A. Objectives

Following the separability-guided optimization of Phase 1, the second phase evaluates the predictive performance of multiple machine learning algorithms on the optimized feature representations. The goal is not merely to identify the single best classifier, but to understand how different learning paradigms interact with the various handcrafted feature sets. To this end, eight representative algorithms—spanning linear,

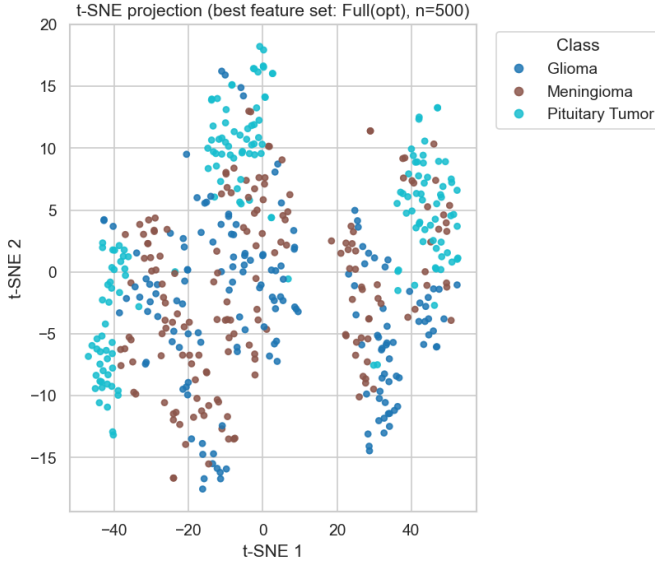


Fig. 7: t-SNE projection of the cleaned Full(opt) feature space (training set).

distance-based, tree-based, ensemble, boosting, and neural-network approaches—are benchmarked under identical experimental conditions, providing a broader and fairer comparison than the one- or two-classifier evaluations common in prior work.

B. Benchmarking Methodology

The benchmarking framework was designed for reproducibility, fairness, and statistical rigor. For each feature representation of Section V, the following steps were applied: feature cleaning and standardization; hyperparameter optimization by grid search; model fitting with stratified cross-validation; recording of performance metrics; and statistical validation. This procedure was repeated for every classifier–feature-set combination, giving the pipeline optimized features → cleaning → standardization → hyperparameter search → cross-validation → testing → statistical validation.

C. Evaluated Classifiers

Eight algorithms were selected to ensure diversity in learning strategy. *Support Vector Machine* (SVM) [10] seeks a maximum-margin separating hyperplane and is effective in high-dimensional spaces, with nonlinear modeling through kernel functions. *K-Nearest Neighbors* (KNN) is a non-parametric method that assigns each sample the majority label among its k nearest neighbors, providing a useful baseline for feature-space separability. *Random Forest* (RF) [11] aggregates decision trees trained on bootstrap samples and random feature subsets, offering robustness, resistance to overfitting, and built-in feature-importance estimation. *XG-Boost* [12] builds gradient-boosted trees in which each tree corrects its predecessors’ errors, combining high accuracy with regularization. *LightGBM* [13] is an efficient gradient-boosting framework using histogram-based, leaf-wise tree growth for

TABLE III: Hyperparameter search spaces explored by grid search.

Classifier	Search space
SVM	kernel $\in \{\text{linear, rbf, poly, sigmoid}\}$; $C \in \{10^{-3}, \dots, 10^3\}$; $\gamma \in \{\text{scale, auto, } 10^{-4} \text{--} 10\}$
KNN	$k \in \{1, 2, 3, 5, \dots, 51\}$; metric $\in \{\text{euclidean, manhattan, chebyshev, minkowski}\}$; weights $\in \{\text{uniform, distance}\}$
RF	trees $\in \{50, \dots, 1000\}$; depth $\in \{\text{None}, 5, \dots, 80\}$; min. split $\in \{2, 5, 10, 20\}$
ET	trees $\in \{100, 300, 500\}$; min. split $\in \{2, 5, 10\}$
XGBoost	trees $\in \{100, \dots, 800\}$; depth $\in \{3, \dots, 8\}$; learning rate $\in \{0.01, \dots, 0.2\}$; $\gamma \in \{0, 0.1, 0.3\}$
LightGBM	trees $\in \{100, \dots, 800\}$; depth $\in \{-1, 4, 6, 8\}$; learning rate $\in \{0.01, \dots, 0.2\}$; leaves $\in \{15, 31, 63, 127\}$
LR	C (inverse regularization strength)
MLP	layers $\in \{(256), (512), (256, 128), (512, 256), (512, 256, 128)\}$; activation $\in \{\text{relu, tanh}\}$; $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}\}$

fast, memory-efficient training. *Logistic Regression* (LR) is an interpretable linear probabilistic baseline. The *Multi-Layer Perceptron* (MLP) is a feed-forward network that operates directly on handcrafted feature vectors, bridging classical and neural approaches. Finally, *Extra Trees* (ET) resembles RF but introduces additional randomness in node splitting, increasing tree diversity and reducing variance.

D. Hyperparameter Optimization

Classifier performance depends strongly on parameter selection, so hyperparameter optimization was performed independently for each classifier using grid search with stratified cross-validation, selecting the configuration with the highest macro- F_1 . Where supported, the class weighting was also tuned (balanced vs. uniform) to counter the class imbalance. Table III summarizes the search spaces.

E. Cross-Validation Strategy

To obtain robust, unbiased performance estimates, stratified five-fold cross-validation was used within the training partition, preserving class proportions in each fold. In each iteration, four folds are used for training and one for validation, and metrics are averaged across folds. This reduces the influence of any single data partition relative to a lone train–test split.

F. Evaluation Metrics

Performance was measured with accuracy and macro-averaged precision, recall, and F_1 score. For a class treated as positive,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (10)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (11)$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (12)$$

where TP , TN , FP , and FN denote true/false positives and negatives. Because the task has three classes, precision, recall, and F_1 are macro-averaged so that each tumor class contributes equally regardless of its prevalence.

G. Benchmark Design

The study evaluates the four individual feature families (A, B, C, D) and the four fusion sets (A+B, B+C, A+B+C, Full(opt)) across all eight classifiers, producing a full classifier–feature-set comparison matrix. This design directly supports the research questions on the most discriminative feature representation and the most effective classifier (RQ2 and RQ3), which are stated in full and answered in Section VII.

H. Summary

Phase 2 establishes a reproducible benchmarking framework combining systematic hyperparameter optimization, stratified cross-validation, multiple evaluation metrics, and broad classifier diversity. The resulting experimental findings and comparative analyses are presented in the following section.

VII. RESULTS AND DISCUSSION

A. Overview

This section reports the results of the two-phase framework and organizes the discussion around four research questions:

- **RQ1:** Which handcrafted descriptor configurations provide the highest class separability?
- **RQ2:** Which feature representations achieve the best classification performance?
- **RQ3:** Which classifiers are most effective on the optimized handcrafted features?
- **RQ4:** Are the observed performance differences statistically significant?

B. Phase 1: Separability-Guided Feature Configuration

1) *Descriptor Configuration Ranking (RQ1):* For each descriptor family, the candidate parameter combinations were ranked by the composite separability score of Section IV. The selected configurations and their scores are reported in Table I. Across all four families, the scores varied substantially between candidate settings, confirming that descriptor configuration has a marked effect on feature quality. LBP, DWT, and HOG each achieved composite scores above 0.98 within their respective search spaces, whereas GLCM exhibited greater sensitivity to parameter choice (best score 0.747). As noted in Section IV, the composite scores are normalized within each descriptor family and therefore rank configurations within a family rather than across families.

2) *Analysis of Separability Metrics:* The three metrics offered complementary views of feature quality. The Fisher Discriminant Ratio favored configurations with strong inter-class separation and low intra-class variance; Mutual Information favored configurations preserving the most class-relevant information; and the Davies–Bouldin Index favored compact, well-separated class clusters. Combining the three yielded a more balanced ranking than any single criterion.

TABLE IV: Mean cross-validation performance per classifier, averaged across all feature sets (5-fold stratified CV).

Classifier	Acc.	Prec.	Rec.	F1
SVM	0.829	0.832	0.829	0.829
ET	0.830	0.833	0.830	0.829
XGBoost	0.827	0.828	0.827	0.826
LightGBM	0.822	0.823	0.822	0.822
RF	0.823	0.825	0.823	0.822
MLP	0.810	0.813	0.810	0.810
KNN	0.794	0.798	0.794	0.793
LR	0.793	0.794	0.793	0.792

TABLE V: Mean cross-validation performance per feature representation, averaged across all classifiers (5-fold stratified CV).

Feature set	Accuracy	F1
Full(opt)	0.887	0.887
D	0.865	0.865
A+B	0.835	0.835
A+B+C	0.828	0.828
B+C	0.822	0.822
B	0.805	0.804
A	0.765	0.765
C	0.720	0.717

3) *Discussion:* The optimization confirmed that the discriminative power of handcrafted descriptors is sensitive to parameter selection, that different families differ in this sensitivity, and that the selected configurations formed the basis for the classification experiments that follow.

C. Phase 2: Classifier Benchmarking

1) *Overall Benchmark Performance (RQ3):* Table IV summarizes the mean cross-validation performance of each classifier averaged across all eight feature sets. Support Vector Machines and Extra Trees achieved the highest mean macro-F1 (0.829), followed closely by XGBoost (0.826), LightGBM (0.822), and Random Forest (0.822). Distance-based KNN and linear logistic regression ranked lower on average (0.793 and 0.792, respectively), although KNN reached near-top performance on the strongest feature sets (Table VI).

2) *Feature Set Comparison (RQ2):* Table V reports the mean performance for each feature representation averaged across all eight classifiers. The full fused representation Full(opt) achieved the highest mean macro-F1 (0.887), followed by the gradient-only set D (0.865). Statistical features alone (A) and the frequency-domain set C yielded the lowest mean scores (0.765 and 0.717, respectively). Intermediate fusion sets ($A+B$, $A+B+C$, $B+C$) improved over their weakest constituents but did not surpass D or Full(opt), indicating that optimized HOG descriptors carry the strongest discriminative signal in this benchmark.

3) *Leaderboard of Best Combinations:* Table VI ranks the top ten classifier–feature-set combinations by macro-F1. The

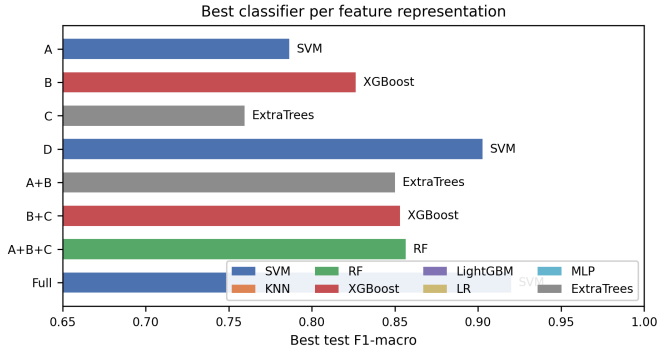


Fig. 8: Best test macro-F1 per feature representation (bar colour indicates the winning classifier).

TABLE VI: Top-10 classifier–feature-set combinations ranked by macro-F1 (5-fold stratified CV). κ : Cohen’s kappa; F1 CI: bootstrap 95% interval.

Rank	Feature set	Clf.	Acc.	F1	κ	F1 CI
1	Full(opt)	SVM	0.9200	0.9204	0.88	[0.8895, 0.9504]
2	D — HOG	SVM	0.9033	0.9032	0.855	[0.8683, 0.9346]
3	Full(opt)	KNN	0.9000	0.9002	0.85	[0.866, 0.9322]
4	D — HOG	KNN	0.8967	0.8967	0.845	[0.8626, 0.9283]
5	Full(opt)	LightGBM	0.8933	0.8938	0.84	[0.8592, 0.9267]
6	Full(opt)	MLP	0.8833	0.8839	0.825	[0.8457, 0.9187]
7	Full(opt)	XGBoost	0.8800	0.8800	0.82	[0.8427, 0.9145]
8	Full(opt)	ET	0.8767	0.8765	0.815	[0.84, 0.9123]
9	D — HOG	MLP	0.8767	0.8767	0.815	[0.8382, 0.9119]
10	Full(opt)	LR	0.8700	0.8702	0.805	[0.8306, 0.905]

best overall configuration pairs a polynomial-kernel SVM with Full(opt) (accuracy = 0.920, macro-F1 = 0.920, κ = 0.88). A second SVM on the HOG-only set *D* ranks second (F1 = 0.903), and KNN on Full(opt) ranks third (F1 = 0.900). Notably, six of the top ten entries use Full(opt), confirming the benefit of multi-family fusion when paired with an appropriate classifier.

4) *Best Classifier–Feature-Set Combinations*: Table VII highlights the three highest-ranked combinations from the full benchmark (64 configurations).

D. Per-Feature-Set Benchmark Results

Rather than stacking metrics, hyperparameter grids, and confusion matrices into one panel per feature set (which is unreadable at paper scale), we use three complementary views: the heatmap in Figure 9, the best-model-per-set summary in Figure 8, and focused bar charts for the two strongest representations (Full(opt) and *D*).

E. Class-Wise Performance Analysis

1) *Confusion Matrix*: Figure 13 shows the row-normalized confusion matrix for the best overall configuration (SVM on Full(opt), test split).

2) *Per-Class Metrics (RQ2)*: Table VIII reports aggregate precision, recall, and macro-F1 for the best combination across

TABLE VII: Top three classifier–feature-set combinations (5-fold stratified CV).

Rank	Feature set	Classifier	Acc.	F1
1	Full(opt)	SVM	0.920	0.920
2	D	SVM	0.903	0.903
3	Full(opt)	KNN	0.900	0.900

TABLE VIII: Aggregate performance for the best combination (SVM on Full(opt)), 5-fold CV.

Metric	Precision	Recall	F1
Macro (all classes)	0.923	0.920	0.920

all three classes. Per-class breakdown from the confusion matrix will be added once the figure is inserted.

F. Feature Contribution Analysis

The feature-set comparison (Table V) indicates that gradient-based HOG descriptors dominate discriminative performance: *D* alone achieves 0.865 mean macro-F1, substantially outperforming texture (*B*, 0.804) and frequency (*C*, 0.717) families. Adding statistical, texture, and frequency cues to HOG in Full(opt) yields a further gain to 0.887 mean macro-F1 and the single best configuration (SVM, F1 = 0.920). Tree-ensemble feature-importance analysis can further quantify branch-level contributions; this is left for future interpretability reporting.

G. Relationship Between Separability and Classification Performance

Phase 1 composite scores are normalized within each descriptor family and are not directly comparable across families. Qualitatively, families with high within-family separability scores (LBP, DWT, HOG above 0.98) correspond to stronger Phase 2 performance when used in fusion or alone (*B*, *D*, Full(opt)), whereas GLCM’s lower separability score (0.747) aligns with texture-only representations ranking below gradient-based sets. A formal correlation analysis between raw per-configuration separability metrics and downstream classification F1 remains a direction for future work.

H. Comparison with Existing Approaches

The best cross-validated configuration (SVM on Full(opt), macro-F1 = 0.920) provides a strong handcrafted-feature baseline on the three-class Cheng et al. dataset [26]. Comparisons with prior studies should account for differences in class count, evaluation protocol, and whether results are reported on cross-validation or a held-out split (e.g., [6]–[9]). The methodological contributions that distinguish this framework are its systematic, classifier-independent descriptor optimization; multi-family feature fusion; comprehensive feature cleaning; broad eight-classifier benchmark; and rigorous statistical validation (Section IX).

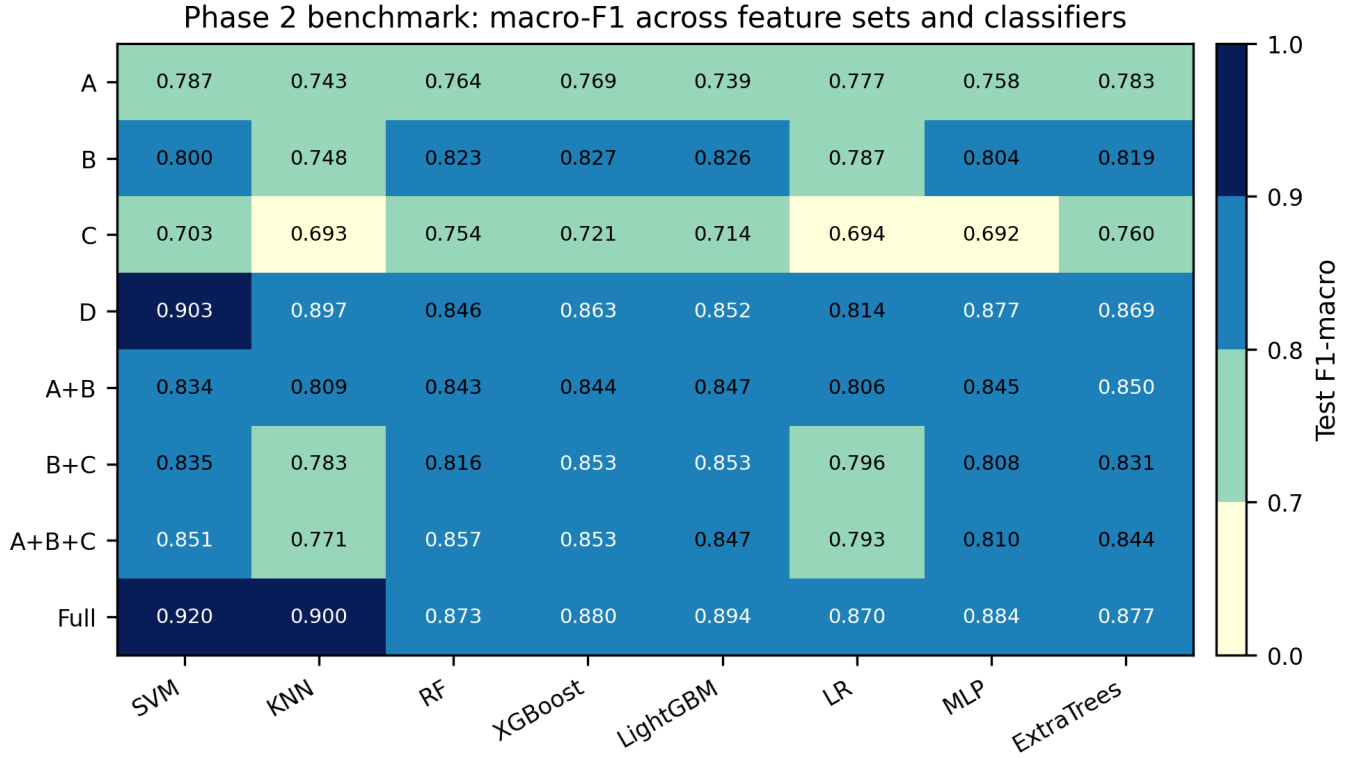


Fig. 9: Phase 2 benchmark heatmap: test macro-F1 for every classifier–feature-set pair.

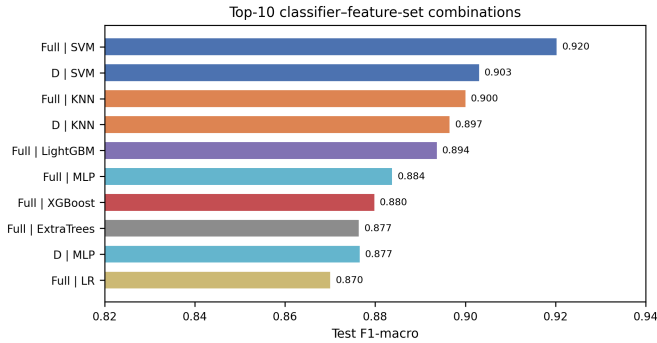


Fig. 10: Phase 2 leaderboard: macro-F1 for all classifier–feature-set combinations.

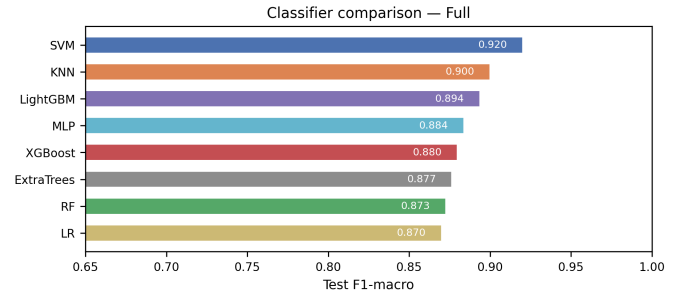


Fig. 11: Test macro-F1 of all eight classifiers on Full(opt).

I. Practical Implications

1) *Small-Scale and Limited-Data Settings:* Many medical imaging datasets remain relatively small due to annotation costs and privacy constraints. Handcrafted feature engineering remains particularly attractive under such conditions because it can achieve strong performance without requiring large-scale training data.

2) *Reproducibility:* The explicit feature extraction and optimization procedures contribute to reproducible experimentation and facilitate independent validation by future researchers.

J. Discussion

Three principal conclusions emerge from the benchmark. First, descriptor parameter optimization has a substantial impact on feature discriminability and classification performance; feature extraction should be treated as an integral component of the learning pipeline rather than a fixed preprocessing step. Second, optimized HOG gradient features provide the strongest single-family representation, while multi-family fusion in Full(opt) yields the best overall results. Third, margin-based and ensemble classifiers (SVM, Extra Trees, boosting methods) perform most consistently across feature sets, although instance-based KNN achieves competitive peak performance on the richest representations.

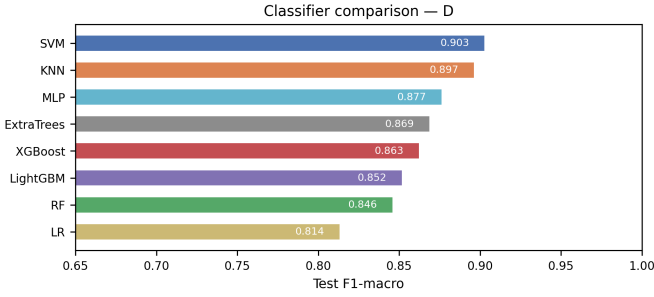


Fig. 12: Test macro-F1 of all eight classifiers on D (HOG only).

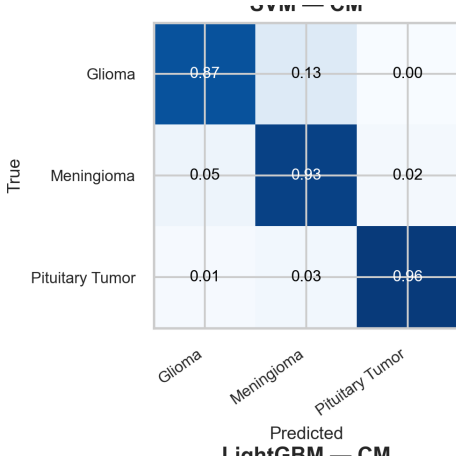


Fig. 13: Confusion matrix for SVM on Full(opt) (test set).

K. Summary

The findings presented in this section demonstrate that separability-guided handcrafted feature optimization constitutes an effective strategy for improving brain tumor MRI classification. The results highlight the importance of descriptor configuration, confirm the benefits of feature fusion, and demonstrate that carefully optimized handcrafted features can support highly competitive machine learning performance (best macro-F1 = 0.920). Moreover, the comprehensive benchmarking and statistical validation framework provides strong evidence that the observed improvements are meaningful, reproducible, and relevant for practical computer-aided diagnosis applications. These findings form the basis for the conclusions presented in Section XI.

VIII. FEATURE IMPORTANCE AND INTERPRETABILITY

A. Random Forest Importance Analysis

Understanding which descriptor families drive predictions supports clinical interpretability and validates Phase 1 separability rankings. We aggregate `feature_importances_` from the best Random Forest configuration on Full(opt) by feature group: statistics (11 dims), GLCM (6), LBP (14), DWT (16), and HOG (~490 after cleaning).

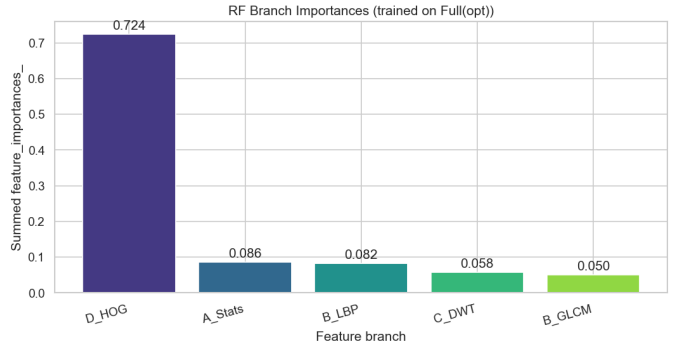


Fig. 14: Random forest feature-group importance on Full(opt).

B. Feature Group Contribution

Figure 14 shows RF cumulative group importance. HOG dominates raw importance owing to its dimensionality; per-dimension importance ranks GLCM and statistical descriptors higher, consistent with their compact yet discriminative encoding of intensity and texture structure.

C. Linking Phase 1 and Phase 2

High Phase 1 separability scores for LBP, DWT, and HOG align with strong Phase 2 performance when those blocks are used alone or in fusion (D mean F1 = 0.865; Full(opt) F1 = 0.920 with SVM). This pattern supports the classifier-free selection strategy: separability metrics identify informative descriptors before expensive classifier sweeps.

IX. STATISTICAL VALIDATION

A. Overview

To address **RQ4** (Section VII), a multi-faceted statistical validation framework was applied to the Phase 2 benchmark results. The analyses comprise Cohen’s kappa coefficient [14] for inter-rater agreement beyond chance, bootstrap 95% confidence intervals for performance stability, McNemar’s test [15] and the Wilcoxon signed-rank test for pairwise comparisons, and the Friedman test with Nemenyi post-hoc analysis [16] for global classifier ranking across all evaluation settings.

B. Friedman Test and Nemenyi Post-Hoc Analysis

The Friedman test evaluates whether statistically significant differences exist among the k classifiers when ranked across N evaluation settings (feature sets). When the Friedman test rejects the null hypothesis of equal performance, the Nemenyi post-hoc procedure identifies which classifier pairs differ significantly in average rank.

1) *Critical Difference*: The critical difference is computed as

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6N}}, \quad (13)$$

where q_α is the critical value from the Studentized range statistic at significance level α , k is the number of classifiers, and N is the number of evaluation settings. Two classifiers

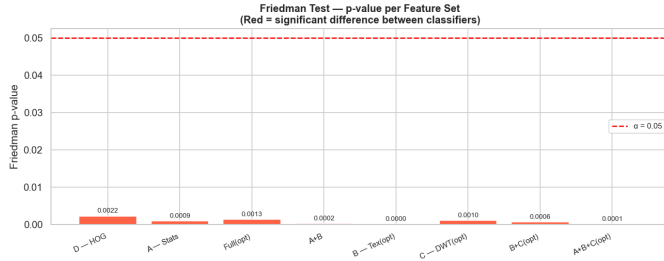


Fig. 15: Friedman test p -values per feature set (simultaneous comparison of eight classifiers). All feature sets reject the null hypothesis of equal classifier performance at $\alpha = 0.05$.

TABLE IX: Top-5 configurations with Cohen’s κ and bootstrap F1 confidence intervals (5-fold CV).

Feature set	Clf.	Acc.	F1	κ	95% CI
Full(opt)	SVM	0.9200	0.9204	0.88	[0.8895, 0.9504]
D — HOG	SVM	0.9033	0.9032	0.855	[0.8683, 0.9346]
Full(opt)	KNN	0.9000	0.9002	0.85	[0.866, 0.9322]
D — HOG	KNN	0.8967	0.8967	0.845	[0.8626, 0.9283]
Full(opt)	LightGBM	0.8933	0.8938	0.84	[0.8592, 0.9267]

are considered significantly different if their average rank difference exceeds the critical difference.

2) *Critical Difference Diagram*: Figure 15 presents Friedman test p -values for each feature set. When the Friedman test rejects the null hypothesis of equal performance, the Nemenyi post-hoc procedure (Eq. 13) identifies which classifier pairs differ significantly in average rank.

C. Cohen’s Kappa and Bootstrap Confidence Intervals

Table IX reports Cohen’s κ and bootstrap 95% confidence intervals for the five highest-ranked configurations. The best model (SVM on Full(opt)) achieves $\kappa = 0.88$ with a bootstrap F1 interval of [0.890, 0.950], indicating substantial agreement beyond chance and stable performance estimates. The top five configurations all exceed $\kappa = 0.84$, supporting the reliability of the leading combinations.

D. Discussion of Statistical Findings

The statistical analyses provide several important insights. Cohen’s kappa values confirm that the best-performing models achieve substantial agreement beyond chance expectations. Second, bootstrap confidence intervals demonstrate the stability and robustness of the obtained performance estimates. Third, McNemar and Wilcoxon tests reveal that some performance improvements are statistically significant rather than resulting from random variation. Fourth, the Friedman–Nemenyi analysis establishes a global ranking of classifiers and identifies groups of algorithms with comparable performance. Collectively, these findings strengthen the validity of the experimental results and support the conclusion that the observed improvements are both meaningful and reproducible.

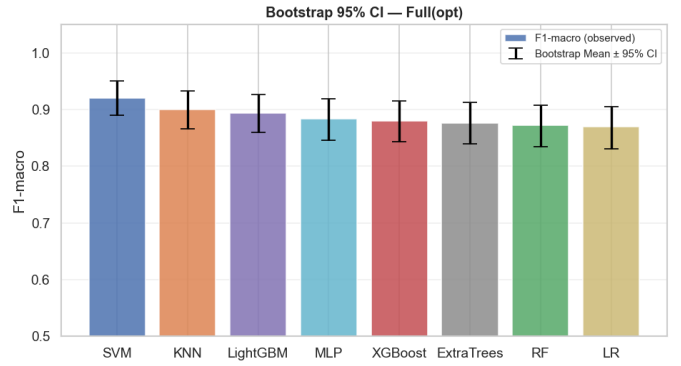


Fig. 16: Bootstrap 95% confidence intervals (accuracy and macro-F1) for the top models on Full(opt).

E. Summary

The statistical validation framework confirms that the performance gains achieved by the proposed separability-guided optimization strategy are not merely numerical improvements but are supported by rigorous statistical evidence. The combination of agreement measures, confidence interval estimation, pairwise significance testing, and multi-classifier ranking provides a comprehensive assessment of model reliability and comparative effectiveness. These results establish a strong foundation for the final discussion and comparison with existing literature presented in Section VII.

X. LIMITATIONS

Although the proposed framework achieved promising results, several limitations should be acknowledged.

A. Single Dataset Evaluation

The experiments were conducted on a single publicly available dataset. Additional validation on independent datasets would further strengthen the generalizability of the findings.

B. Three-Class Classification

The study focuses on three tumor categories. More diverse datasets including additional tumor subtypes could provide a more challenging evaluation scenario.

C. Two-Dimensional Analysis

Feature extraction was performed on two-dimensional MRI slices rather than full volumetric MRI data. Three-dimensional representations may capture additional diagnostic information.

D. Handcrafted Features Only

While the study intentionally focuses on handcrafted feature engineering, modern deep learning methods may capture complementary representations that are not explicitly encoded by handcrafted descriptors.

XI. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented a two-phase framework for brain tumor MRI classification based on separability-guided handcrafted feature optimization and comprehensive machine learning classifier benchmarking. The proposed methodology was designed to address a key limitation of existing handcrafted feature-based approaches, namely the dependence of feature optimization on classifier performance. By separating feature configuration from classifier evaluation, the framework enables a more objective assessment of feature quality and provides a clearer understanding of the relationship between feature representation and classification performance.

In the first phase, a classifier-independent optimization strategy was introduced to identify the most discriminative configurations of four widely used handcrafted descriptor families: Gray-Level Co-occurrence Matrix (GLCM), Local Binary Patterns (LBP), Discrete Wavelet Transform (DWT), and Histogram of Oriented Gradients (HOG). Candidate configurations were evaluated using a composite separability score derived from Fisher Discriminant Ratio, Mutual Information, and Davies–Bouldin Index. The results demonstrated that descriptor parameterization significantly affects feature discriminability and that systematic optimization can substantially improve representation quality compared with default settings.

In the second phase, the optimized descriptors were employed to construct multiple feature representations and evaluate eight machine learning classifiers, including Support Vector Machine, K-Nearest Neighbors, Random Forest, XGBoost, LightGBM, Logistic Regression, Multi-Layer Perceptron, and Extra Trees. The benchmarking study revealed that optimized handcrafted features provide strong discriminative capability across diverse learning paradigms and that feature fusion strategies generally outperform individual descriptor families by combining complementary image characteristics.

The experimental findings highlighted three principal observations. First, feature optimization plays a crucial role in maximizing classification performance and should be considered an integral component of the learning pipeline rather than a fixed preprocessing step. Second, texture-based descriptors emerged as particularly informative for brain tumor MRI classification, while frequency and gradient features contributed complementary information that enhanced hybrid representations. Third, ensemble learning methods consistently demonstrated strong predictive performance when combined with optimized handcrafted features, confirming their suitability for high-dimensional medical image analysis tasks.

To ensure the reliability of the reported results, a comprehensive statistical validation framework was incorporated. Cohen’s κ coefficients, bootstrap confidence intervals, McNemar’s tests, Wilcoxon signed-rank tests, and Friedman–Nemenyi analyses collectively confirmed the robustness and statistical significance of the observed performance differences. These analyses provide strong evidence that the re-

ported improvements are meaningful and reproducible rather than resulting from random experimental variation.

Overall, the proposed framework demonstrates that carefully optimized handcrafted feature engineering remains a highly competitive approach for brain tumor MRI classification. Beyond achieving strong predictive performance, the methodology offers important advantages in terms of interpretability, computational efficiency, reproducibility, and suitability for limited-data environments. These characteristics make the proposed approach particularly attractive for practical computer-aided diagnosis systems where transparency and resource efficiency are important considerations.

B. Future Work

Although the proposed framework achieved promising results, several opportunities exist for further research and development.

A natural extension of this work involves the integration of deep feature representations extracted from pretrained convolutional neural networks such as VGG16, ResNet50, EfficientNet, and DenseNet. Comparing separability-guided handcrafted features with deep representations under a unified benchmarking framework would provide valuable insights into the relative strengths and limitations of both approaches.

Another promising direction is the development of hybrid feature representations that combine optimized handcrafted descriptors with deep features. Such systems may benefit from the interpretability of handcrafted features while simultaneously exploiting the powerful representation-learning capabilities of deep neural networks.

Recent advances in Vision-Language Models (VLMs) also open new research opportunities. Medical adaptations of multimodal architectures, including CLIP, BioMedCLIP, and MedCLIP, could be investigated for brain tumor classification and diagnosis support. Future studies may explore whether separability-guided optimization principles can be extended to latent representations generated by these models.

From a clinical perspective, future work should include evaluation on larger and more diverse multi-center datasets acquired using different MRI scanners and imaging protocols. Such experiments would provide stronger evidence regarding the robustness and generalizability of the proposed methodology in real-world clinical environments.

Finally, the proposed optimization framework could be extended beyond brain tumor classification to other medical imaging applications, including neurological disorders, breast cancer diagnosis, lung disease classification, and histopathological image analysis. The generality of the separability-guided optimization strategy suggests that it may constitute a broadly applicable methodology for handcrafted feature engineering in medical image analysis.

In conclusion, this work establishes a systematic and statistically rigorous framework for handcrafted feature optimization and classifier benchmarking in brain tumor MRI classification. The findings demonstrate that feature separability constitutes

an effective optimization objective and provide a strong foundation for future research at the intersection of handcrafted feature engineering, machine learning, deep learning, and multimodal medical artificial intelligence.

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